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Review

Grain storage systems and effects of moisture, temperature and time on grain quality - A review

Valmor Ziegler [a, *], Ricardo Tadeu Paraginski [b], Cristiano Dietrich Ferreira [a]

[a] Technological Institute in Food for Health, University of Vale do Rio dos Sinos, 93022-750, São Leopoldo, RS, Brazil
[b] Professor do Instituto Federal de Educação, Ciência e Tecnologia Farroupilha, Campus Santo Augusto, Brazil

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A B S T R A C T

Global grain production has been increasing every year, which highlights the need for investment, maintenance, and effective management in warehouses. Warehousing is a major bottleneck as it influences the possibility of negotiating for better market prices. In order for profitability to be increased, assurance of product quality for the consumer is necessary. Hence, there is a need for standardization of scientific research and the production chain in this field. The purpose of this bibliographic review is to describe the existing storage systems for grains, steps to be taken for care, and the main studies carried out in this field, to guarantee grains of good quality. The maintenance of the characteristics of the grains after harvest, is essential to guarantee grains of good quality to consumers. However, there are still deficiencies in the field of grain storage, and features of the different systems and conditions to be adopted for each of them are provided in this review.

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* Corresponding author.
E-mail addresses: valmorziegler12@unisinos.br (V. Ziegler), ricardo.paraginski@iffarroupilha.edu.br (R.T. Paraginski), cristianodietrich@unisinos.br (C.D. Ferreira).

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1. Introduction

Global grain production has been increasing every year, which highlights the need for investment in components of postharvest system. Product prices and options for raw material holders fluctuate over the years, as a result of public policies, government stocks, and trade barriers. In order to allow commercialization after harvest, the grains need to be stored, and their quality needs to be preserved throughout this period. However, with the increase in production levels, there is increasingly a deficit in static storage capacity, which increasing cost and end losses along the marketing chain. The Food and Agriculture Organization of the United Nations (FAO) recommends that a country must have a static storage capacity greater than its production capacity, which guarantees a nation's food security (FAO, 2019). The storage deficit generally leads to qualitative and quantitative losses (CONAB, 2020; FAO, 2019). Allied to these factors, some of the units currently used have a deficient storage structure, and characteristics that hinder correct storage. Basic precepts and systems need to be established to ensure correct storage of grain throughout the process, and these will be discussed throughout this review.

This review seeks to gather information about grain storage, and provide a reference material for future studies, which is a currently observed shortcoming. The review will describe the basic principles of grain storage, systems available for use, a summary of the main storage studies carried out across the world with the major grain species, and the effects of moisture, temperature and time on grain storage.

2. Basic principles of grain storage

The global grain production evolves every year, requiring an increase in the level of technicalization throughout the production process, from pre-sowing to harvest. The grains can be harvested when they reach physiological maturation, but this process is restricted, as it does not allow mechanical threshing. Thus, the grains remain in the crops until they reach an adequate moisture content to allow mechanical threshing. In this time between maturation and harvesting, called the critical period, the grains remain exposed to weather conditions, such as rain, temperature fluctuations, and pest attacks, due to which their quality may be altered. The first measure to be adopted in this context is to reduce the time between maturation and harvesting, which must be carried out as soon as the crops reach the moisture content suitable for mechanical harvesting (Mares and Mrva, 2008).

Among the losses observed during the storage of grains, the quantitative ones are represented by the loss of weight and moisture content of the grains, caused by the grains' respiration process and search for hygroscopic balance, respectively. In terms of qualitative losses, there is loss in taste, odor, nutritional value, metabolites, physiological parameters (germination and vigor), and increased risk of mycotoxins, in addition to market devaluation. Throughout the crop cycle, the crops must be protected from attack by insects, fungi, and mites, which can reduce the nutritional, technological, and bioactive quality of the crops, even before harvest. Climatic conditions, such as rainfall and temperature fluctuations (Mohapatra et al., 2017; Neme and Mohammed, 2017; Mutungi et al., 2019) can also affect the grain quality.

As soon as these grains enter the storage units, it is necessary to review them and assess their quality, to know which processes need to be carried out. Among the many ways the grains enter the units, they can arrive: (1) dry and dirty, (2) dry and clean and (3) moist and dirty. Thus, adequate postharvest steps are required to maintain the quality of the grains, and several measures must be adopted in the grain storage and processing units, before storage (Neme and Mohammed, 2017).

The evaluation of quality of the grains at the time of reception or dispatch from the storage units is essential to determine their market value, and also to decide which postharvest maintenance activities, such as pre-cleaning, cleaning, and drying will need to be performed. Thus, at the time of receipt at the storage units, correct grain sampling, homogenization, and evaluation procedures need to be conducted, along with collecting information, such as cultivar, place of production, collection system, and harvest time, all of which facilitate decision-making inside the storage unit. We must use a composite or pneumatic sampler, always collecting from the minimum number of established sampling points, according to the legislation of each country. The collection must be carried out covering the entire depth of the grain mass and, these samples must be homogenized in Boerner-type or Jones-type homogenizers after collection. Later, the samples must be quartered in the same equipment to obtain working samples. Compulsory analyses should be carried out to determine moisture content, foreign matter, and impurities; to identify defects, and establish final typification, all of which must be described in a Standard Operating Procedure (SOP).

The evaluation of grains carried out at reception in a storage unit can determine if these grains need to undergo pre-cleaning and cleaning steps. Cleaning is characterized as the process of removing foreign matter and impurities from the grain mass, and needs to be carried out before storage to standardize the grain mass (Guedes et al., 2011). When drying grains, the levels of foreign matter and impurities are reduced to values below 4%. For storage, it is recommended that these values should be below 1%. The pre-cleaning and cleaning steps must be carried out with air machines, which use the principle of air flow, and with sieves of different sizes to obtain the desired level of cleaning. The presence of foreign matter in the grain samples at the time of drying, can cause fire, and put the entire storage unit at risk, depending on how long they remain inside silos. The permanence of grains inside the silos, on the other hand, can favor the development of microorganisms, and also hinder the aeration process of the silos (Jian et al., 2019).

Drying is a fundamental step in the conservation of all grains, and is characterized as the process of removing water from the grains to levels that allow storage for long periods. Moisture is the main factor that alters grain properties and makes the environment favorable for the development of insects and microorganisms (Coradi and Lemes, 2018; Mohapatra et al., 2017; Paraginski et al., 2015). The permitted water content in grains for storage depends on the species and the period for which the grains are to be stored. In addition to moisture content of the grains, factors such as temperature, relative humidity, pests, and diseases can interfere with the quality of the grains.

Among the drying methods available for grains, we can adopt natural, adapted, and technical methods, based on requirement. However, as most grains are produced on a large scale, technologically advanced drying systems, including stationary, intermittent, continuous, staggered, and dry-aeration systems, are used. A parameter that must be considered before drying, is the reduction of waiting time before the grains are dried, as the grains often reach the storage unit late, and due to a high inflow of product, they end up remaining in the hopper, where, in most cases, the grains are heated. In the short term, this heating does not pose any problems, but over the course of storage, these grains that have undergone heating before the drying process, tend to develop a greater amount of metabolic defects, which downgrades the final typification of the product and may compromise its marketing value. Thus, it is recommended that, while waiting for the drying process, the grains remain inside lung silos that have a natural or artificial cooling system, to reduce the temperature of the grain mass to about 15 °C, with the aim of reducing metabolic activity and changes in these grains (Ferreira et al., 2019; Lang et al., 2020a, 2020b; Timm et al., 2020).

The process of loading silos and warehouses is essential for uniformity in the mass of the grains. The accumulation of foreign matter, impurities, and broken grains at specific points inside the silo should be avoided, as it can hinder the aeration of the grains. Thus, it is recommended to load the silo to one-third of its capacity, remove the grains from the central part of the grain mass, load up the silo again to two-third of its capacity, remove the grains from the central part of the mass again, carry out total loading of the silo, remove the grains from the central part of mass again, and finally carry out the full loading of the silo by adding the grains removed, and keeping a distance of 1.2–1.5 m between the surface of the grain mass and the silo cover. Maintaining this distance from the silo cover is essential to facilitate gas exchange in the space, and to prevent condensation from occurring on the surface of the grain mass. The use of exhaust fans is essential to assist in gas exchange and prevent condensation from occurring (Jian et al., 2019).

3. Grain storage

When storing grains, it is also necessary to conserve them, and this requires care, knowledge, dedication, and professionalism from the individuals involved in the process. The ability to preserve the quality, health, and nutritional value of the grains during the storage period, does not depend only on the conditions of production and harvest, but also on storage, and maintenance of appropriate conditions for storage (Mobolade et al., 2019).

Grains, despite morphological characteristics of resistance and sturdiness specific to each species, are subject to attacks by insects, mites, microorganisms, rodents, birds, and other animals. Mechanical damage, biochemical changes, and effect of non-enzymatic chemicals is also seen, even before storage of the grains. This set of undesirable factors causes quantitative and/or qualitative losses, due to changes in the chemical composition, reduction in the nutritional value, development of toxic substances in the grains, consumption of energy reserves, and a decrease in their commercial value. Consequently, these factors end up compromising the use of the grains for consumption and even for industrialization, if adequate techniques and efficient conservation methods are not adopted (Mobolade et al., 2019; Mohapatra et al., 2017; Neme and Mohammed, 2017).

In grains destined for storage, factors such as biological integrity, physical integrity, health status, purity, and moisture must be considered. Changes that occur during storage are reflected in quantitative and/or qualitative losses. Quantitative losses are the most easily observed, reflecting the metabolism of grains and/or associated organisms, and resulting in a reduction in the dry matter content of the grains. Qualitative losses, on the other hand, are mainly due to chemical and enzymatic reactions, presence of foreign matter, impurities, and microbial attack. This results in losses of nutritional, germinative, and commercial value, with the possibility of the formation of toxic substances in the stored grains, if the storage process is not properly conducted (Paraginski et al., 2014; Fleurat-Lessard, 2017; Demito et al., 2019).

Storage systems are grouped according to practical application, into non-hermetic (conventional), semi-hermetic or bulk (horizontal silos, vertical silos and warehouses), hermetic, refrigerated, and modified atmosphere storage.

3.1. Conventional system

The non-hermetic system, also known as the conventional system, is the most diversified storage system, ranging from rustic units, such as storerooms, and barns, to larger and more technicalized units, such as conventional warehouses. These systems generally use sacks made with different materials and of different sizes for storage. The use of larger packages that store up to 1000 kg of grains, has been increasing in places with higher production, and they are easily transported with the aid of mechanical equipment developed by agricultural companies (Mobolade et al., 2019).

The conventional storage system allows the storage of several grains in the same warehouse, and higher quality grains can be separated from lower quality grains during the storage, a facility that bulk storage systems do not allow. However, for the grains to be stored in this system, moisture needs to be reduced to levels below the moisture traditionally indicated for storage, due to the greater area of possible water and thermal exchanges with the environment during storage.

Despite the advantages described above, this system requires packaging, which can be reused in some cases, in addition to pallets, which need to be placed on the surface of the warehouse floors, to avoid direct contact of the grains with the floor, as very humid environments can cause the grains to reheat, and consequently reduce the quality of the grains at the end of the storage period. In addition, the piles of grains cannot exceed dimensions of 19 × 19 m at the base, and must be spaced apart from each other, and from the wall by 0.5 m. A central corridor that allows the passage of mechanical equipment for transportation, is also needed. Finally, considering that the height of the pallets is approximately 25 cm above the floor, these restrictions result in a reduction of approximately 20% of the storage space, when compared to a bulk storage system. Conventional warehouses are used for storage in greater quantities than in storerooms and barns. They are mostly used by cereal farmers and cooperatives that store several grain species simultaneously in the same unit.

The grains are placed in bags after cleaning and drying, and remain there until commercialization or consumption. When pests such as insects, generally introduced during sampling, develop inside these bags, purging of the grains inside the bags must be carried out using phosphine supplied from metal nozzles. All bags are protected with the help of high density plastic canvas, and “sand snakes” are used to seal enclosures used to facilitate fumigation, hence preventing phosphine gas (PH_3) from being lost to the environment after the purging, which would cause a reduction in the efficacy of the fumigation.

3.2. Semi-hermetic or bulk system

The semi-hermetic storage system, also called the bulk system, is a widely used system of grain storage, being characterized as a system that allows gas exchange between the interior of the grain mass and the environment. However, these exchanges are smaller when compared to non-hermetic systems, and larger when compared to hermetic systems (Coradi et al., 2020). Bulk storage does not allow for individualization of cargo or direct inspection, making sample collection difficult. Maintenance or quality control measures, such as temperature control, are carried out indirectly, such as by thermometry. Cable systems with thermocouples, made of constantan and copper alloy, predominate in thermometry (Tefera et al., 2011).

The semi-hermetic storage system allows the storage of large quantities, and easy control of the internal conditions of the grain mass. This is a highly technicalized system, with low maintenance costs, as it does not require bags and pallets. However, it only allows the storage of a single species of grain. In units of the bulk storage system, unlike conventional ones, there is no need for internal space for the movement of people, machines, and/or equipment, but an unfilled space at the top, corresponding to 1.5 m from the upper ceiling of the silo, is required. This space serves for inspection, acts as a reserve for volumetric expansion of the load, and is used for thermal damping.

This space of at least 1.5 m between the surface of the grain mass and the silo cover is required, because on hot days, and especially in the hottest hours of the day, the silo walls and cover receive heat and the temperature inside the silo increases. If there is no free space between the ceiling and the surface of the grain mass, or even if the distance between them is less than 1.5 m, temperature fluctuations during the day can result in water condensation on the ceiling, compromising the quality of the grains. Further, the installation of exhaust fans on the walls of silos and warehouses helps in preventing the occurrence of this unwanted phenomenon.

Vertical storage silos, and bulk warehouses are the main examples of this storage system. In this system, forced ventilation with unheated ambient air through the action of fans, in a process known as aeration, helps in the conservation of pre-dried grains. In other cases, the conservation is by transillation, where there is total movement of the grains stored in a unit to another, or intrasilage, where there is partial movement of the grains within the same unit. Internal conditions of the grain mass can be measured by thermometry systems.

Aeration can be defined as the practice of ventilating the grains with scientifically regulated air flow, to promote reduction in and uniformity of the temperature in the stored grain mass. The aim is to conserve the grains, by reducing the metabolic activities of the grains and of associated organisms. Aeration is necessary to prevent the formation of convective air currents, and to inhibit anaerobiosis within the grain mass. When the temperature outside the silo is higher, or when there is an increase in temperature inside the silo on hot days, the grains near the walls heat up more than the grains towards the center of the mass, as does the air that is close to the wall, causing its density to decrease and upward currents of air to be formed close to the wall, thereby causing air molecules in the center of the silo to form a downward stream of cold air (Lopes et al., 2006).

The air molecules that circulate are unsaturated (they have the capacity to receive water molecules) and when they pass through warmer regions, they absorb heat and their enthalpy increases, thereby increasing their capacity to exchange thermal energy with the water molecules of the grains they pass through. After a period of convective currents, in the central region of the lower third part of the silo, at the coldest point of the grain mass, water condensation occurs, and reaches the dew point. The grains located in this region humidify, and suffer deterioration in quality. Similarly, when the ambient temperature is lower (cold hours and/or days), the air near the silo wall cools and forms a downward convective current, causing an upward convective current of air at the center of the mass of grains. This results in a condensation region at the top of the silo, since the silo cover is cold, and a condensation zone is formed in the cone region of the silo. In temperate climates, heat transfer phenomena happen every day (Lopes and Neto, 2019).

Periodic aeration is essential to prevent these changes in air temperature in the silo from altering the characteristics and quality of the grain mass. Fans, which need to be correctly sized for each type of silo, allow the air to reach all parts of the silo, with adequate air flow and static pressure. This operation requires time, and needs to be carried out when the air has adequate psychrometric conditions, such as low temperature and relative humidity.

The flux of hot and humid air into the grain mass can cause heating, high temperature, and high humidity over time, which accelerates the respiratory process of the grains and facilitates the development of insects and fungi in the grain mass. This can compromise the quality of the stored grains (Lopes and Neto, 2019; Lopes et al., 2006). Air cooling equipment increases safe aeration time, and thermometry systems help to diagnose temperature changes inside the grain mass.

In addition to forced aeration, eolic exhausters have been used, positioned at the top of the silo, where the wind itself and the pressure differences between the inside and outside of the silo, rotate the fins that suck the air from inside the silo (Da Silva et al., 2014). In addition to aeration, other ways of promoting grain ventilation are transillation and intrasillation. In the first, there is total transfer of all grains from one silo to another, or from one cell to another, in the case of septate bulk warehouses. In the second, there is passage of part of the grains through an elevator, with return to the same cell or the same silo. In aeration, the air is forced through the mass of grains with the help of a fan, depending on the system, while in transillation and intrasillation, the grains pass through air, with the aid of an elevator.

Transillation is a practice that must be performed during the loading of the silos, especially in the vertical ones. When the height of the silos is high, it increases the pressure on, and consequently the damage to the grains. Thus, when the loaded grain mass reaches one-third of the maximum height of the silo, the grains from the central part of the mass must be removed, where mainly broken grains and impurities are concentrated. The grains removed must be cleaned before they are placed on the top of the grain mass again. Afterwards, the loading is continued, and when it reaches two-thirds of the height of the silo, another transillation is carried out, removing the grains from the central part of the grain mass. The loading is further continued, and when the loading of the upper third part of the silo is completed, a new transillation step is carried out, and the grains removed are placed back on top of the grain mass in the silo. This practice can be performed in vertical silos, and bulk warehouses.

The removal of broken grains, foreign matter and impurities, which accumulate in the center of silos and warehouses, assists in the aeration process, and in the conservation of the grains. During aeration, the presence of broken grains increases the static pressure of the air, which is necessary for the air to reach the surface of the grains. The air always seeks regions where the resistance is less, and thus does not pass through the regions where the concentration of the broken grains is high. In addition, broken grains are more susceptible to the development of fungi and attack by insects, because physical integrity of the grains is compromised. The respiration process is higher in broken grains, which increases the breakdown of dry matter of the grains, by lipid, protein, and carbohydrate degradation, resulting in a lower weight of the grains, which is incorrectly termed as “technical breakdown.” However, with adequate storage processes and techniques, these losses can be eliminated (Jian et al., 2019).

3.3. Hermetic system

The hermetic system, does not apply for large volumes. Casks, drums, and other containers, for small quantities, and “plastic silos” for slightly larger quantities predominate in this system, while in emergencies, the most used models are “the sack pools.” However, in recent years, emergency warehouses have become a storage alternative for producers.

These hermetic storage systems for dry or wet grains are based on the principle of reducing the oxygen available in the storage ecosystem to lethal or limiting levels for the associated living organisms. This reduction can be achieved spontaneously through the respiratory process of the existing grains and organisms, by exhaustion of air, and by burning candles, cotton and other materials inside the container, which reduces the level of oxygen (O_2) and increases the level of carbon dioxide (CO_2). CO_2 is one of the final products of metabolism of organic substrates in living organisms, and has a conservative effect in hermetic conditions, with inhibitory action on the enzymatic activity of the grains and associated organisms, thereby causing death of the microorganisms (Govereh et al., 2019; Odjo et al., 2020).

Hermetic grain storage conditions act selectively on insect populations by decreasing their activity, inhibiting and/or biological paralysis (Odjo et al., 2020). The degree of moisture in the grains, storage time, temperature of the inter-granular atmosphere, specific resistance characteristics of the species of grains, and different stages of development of the grains are the main factors that influence the efficiency of the hermetic storage system. The incomplete metabolism of carbohydrates in the grains through their respiratory process, and by the associated organisms under restricted aerobic and/or anaerobic conditions, together with the presence of bacteria and yeasts, leads to the formation of ethyl alcohol, and short-chain organic acids, like acetic acid, lactic acid, and butyric acid. These products have a secondary conservative effect, being able to alter the sensory characteristics of the stored grains, such as odor and flavor, which are not always removed by the end of the storage process (Baributsa and Njoroge et al., 2020). The partial hydrolysis of certain nutrients in the grains, such as carbohydrates and proteins, into simple sugars and amino acids respectively, in readily assimilable forms, can represent a nutritional advantage in animal feed store hermetically.

The production of CO_2 during the storage of grains under hermetic conditions, results in a considerable excess of internal pressure in the storage structures, which is directly proportional to the degree of moisture of the stored grains. This aspect is an important technical parameter to be considered when planning hermetic structures for the storage of grains. Obtaining and maintaining airtightness are the main technical aspects for the efficiency of the system, since a simple space that allows gas exchange, compromises the efficiency of the system, and consequently, results in low quality grains at the end of the storage period, especially a long one (Baributsa and Njoroge, 2020).

This type of storage system is a good alternative for small producers, especially family farmers, as they can use plastic packaging (PET), which has been discarded after use for recycling. These packages must be washed and dried, after their original products, mainly soft drinks, have been consumed. Later, the packages are filled completely with the grains, in order to allow the least amount of air to remain inside. During the respiratory process of the grains, the oxygen present will be consumed, resulting in an airtight environment, which improves the preservation of the color and characteristics of the grains during storage and before consumption, when compared to the conventional system of storage in sacks (Baributsa and Njoroge, 2020; Scariot et al., 2018; Odjo et al., 2020; Mlambo et al., 2017).

3.4. Refrigeration system

The use of artificial cooling technology for grains has been increasing every year, being one of the main technologies developed for grain storage. The technology consists of reducing the temperature of the grain mass, from about 25 °C to about 15 °C, thus reducing changes in the grains by their respiratory process, and also reducing their susceptibility to insect attack, because in temperatures below 17 °C, the development rate of insects drops close to zero (Ziegler et al., 2016; Paraginski et al., 2014; Demito et al., 2019; Ferreira et al., 2018).

The temperature reduction is achieved with artificial equipment, which alters the properties of the air, and reduces the temperature and absolute humidity of the air, before it comes in contact with the grain mass. Thus, cold and dry air is blown into the grain mass, mainly in silos and vertical warehouses. This process is slow, because as it occurs in aeration, there is a cooling front, thus, all the dough needs to be refrigerated, which can vary depending on the capacity and especially the height of the silos.

Producers wishing to use this technology need not necessarily purchase the machine for refrigeration, as there are companies that currently provide these services. As grains are good insulators, they can be refrigerated, and the low temperature will be maintained for a long period, which depends on the psychrometric conditions of the air in the storage unit. However, constant refrigeration is not necessary. Hence, a silo can be cooled in one place, and the refrigeration equipment can be transferred to another place, returning after a certain period of time for a repeated cooling of the grains. The increase and variations in the temperature of the grain mass can be observed with the aid of thermometry systems.

3.5. Modified atmosphere system

An alternative for producers is the storage of grains in a modified atmosphere, a technology already used by many producers, which helps in maintaining the quality of the grains, and obtaining higher market values in the off-season than those of grains traditionally stored in conventional systems (Rupollo et al., 2011; Lindemann et al., 2019). Some companies already specialize in the construction of metal silos with less capacity than those traditionally built, usually with a capacity of 300000 kg, which allows better control of the behavior of bean grains during the period of storage.

The use of this technique provides better quality grains at the end of storage, with higher market value, helps in reducing the darkening and hardening of the grains, and prevents the development of insects within the grain mass. It also helps to improve the physicochemical properties of the grains after cooking, such as a higher total solid content in broth, greater protein solubility, and lesser grain texture than conventionally stored grains. These are characteristics that increase the commercial value of the grains, and consequently, the income of the producer at the time of commercialization (Rupollo et al., 2011; Lindemann et al., 2019).

The modified atmosphere storage system consists of modifying the traditional dry or wet grain storage atmosphere, reducing the oxygen levels available in the storage ecosystem to lethal or limiting levels for associated living organisms, and also reducing the grain's respiratory process. These conditions are obtained forcibly with the use of nitrogen (N_2) and/or CO_2, along with the suppression of O_2 (Jayas and Jeyamkondan, 2002; Moncini et al., 2020).

The use of nitrogen in place of oxygen is the most indicated, because high levels of carbon dioxide can put workers' lives at risk due to its high toxicity. Nitrogen gas can be easily purchased from retailers, and companies currently specialize in making these available to consumers. An alternative to the use of this technology by producers is the use of polyethylene bags developed by companies. These bags do not have gas permeability, have a higher thickness, and are packaged by machines. The machines remove oxygen and add nitrogen into the packages, ensuring the maintenance of the modified atmosphere during storage.

4. Changes in grain quality due to storage

Table 1 depicts studies carried out since 2008 on the storage of cereals, beans, and oil seeds. These studies have mainly focused on variables such as moisture content in grain, temperature, relative humidity, and storage time, which are also the focus points of our review.

Table 1
Main studies on the storage of cereals, beans and oilseeds after 2008.

Species	Storage conditions			Article
	Moisture	Time	Temperature	
Black bean	11.5% and 11.6%	225 days	20 °C	Physicochemical properties and enzymatic bean grains dried at different temperatures and stored for 225 days (Elias et al., 2016)
Black bean	14% and 17%	12 months	11, 18, 25 and 32 °C	Characteristics of starch isolated from black beans (Phaseolus vulgaris L.) stored for 12 months at diferente moisture contents and temperatures (Ferreira et al., 2016)
Black bean	14% and 17%	12 months	11, 18, 25 and 32 °C	Physicochemical, antioxidant and cooking quality properties of long-term stored black beans: effects of moisture content and storage temperature (Ferreira et al., 2017)
Black bean	14% and 17%	12 months	11, 18, 25 and 32 °C	Quality of black beans as a function of long-term storage and moldy development: Chemical and functional properties of flour and isolated protein (Ferreira et al., 2018)
Carioca Bean	16.7%, 13.8% and 12.0%	240 days	20, 28 and 36 °C	Effects of refrigeration on biochemical, digestibility, and technological parameters of carioca beans during storage (Demito et al., 2019)
Carioca Bean	12.5%	360 days	5, 15 and 25 °C	Pasting, morphological, thermal and crystallinity properties of starch isolated from beans stored under different atmospheric conditions (Rupollo et al., 2011)
Pinto beans	12%, 14%, 16%, 18% and 20%	16 weeks	10, 20, 30 and 40 °C	Storage studies on pinto beans under different moisture contents and temperature regimes (Rani et al., 2013)
Wheat	14.5%	240 days	15 and 30 °C	Influence of temperature and r.h. during storage on wheat bread making quality (González-Torralba et al., 2013).
Wheat	–	10 years	- 18 and 20 °C	Comparative physiology and proteomics of two wheat genotypes differing in seed storage tolerance (Chen et al., 2018)
Wheat	10.6% and 10.78%	270 days	45 °C	Effect of artificial aging on wheat quality deterioration during storage (Tian et al., 2019)
Wheat	12%	1 year	25 and 50 °C	Physiochemical changes in wheat of different hardnesses during storage (Zhang et al., 2017)
Wheat	10.90% and 11.23%	180 days	–	Influence of storage conditions on the quality properties of wheat varieties (Kibar, 2015)
Wheat	–	180 days	18–27 °C and 4-6 °C	Retention of deoxynivalenol and its derivatives during storage of wheat grain and flour (Zhang et al., 2016)
Soy	12%, 15% and 18%	12 months	11, 18, 25 and 32 °C	Changes in the Bioactive Compounds Content of Soybean as a Function of Grain Moisture Content and Temperature during Long-Term Storage (Ziegler et al., 2016)
Soy		2 years	Environment	Changes occurring in compositional components of black soybeans maintained at room temperature for different storage periods (Lee and Cho, 2012)
Soy	12%	8 and 12 months	4, 20 and 30 °C	Functional properties of protein isolates from soybeans stored under various conditions (Liu et al., 2008)
Soy	12% and 15%	12 months	11, 18, 25 and 32 °C	Effects of moisture and temperature during grain storage on the functional properties and isoflavone profile of soy protein concentrate (Ziegler et al., 2018)
Soy	11.2% 12.8% and 14.8%	180 days	20, 30 and 40 °C	Quality of soy bean grains stored under different conditions (Alencar et al., 2009)
Soy	9%, 11% and 13%	12 months	10, 20 and 30 °C	Soybean grain storage adversely affects grain testa color, texture and cooking quality (Yousif, 2014)
Soy	12% and 16%	12 months	15–25 °C	Effects of temperature and moisture during semi-hermetic storage on the quality evaluation parameters of soybean grain and oil (Ziegler et al., 2016b).
Rice	16.5%	4 months	4, 20, 30 and 40 °C	Changes in physicochemical characteristics of rice during storage at different temperatures (Park et al., 2012)
Rice	13%	6 months	16, 24, 32 and 40 °C	Changes in properties of starch isolated from whole rice grains with brown, black, and red pericarp after storage at different temperatures (Ziegler et al., 2017b)
Rice	13%	6 months	16, 24, 32 and 40 °C	Cooking quality properties and free and bound phenolics content of brown, black, and red rice grains stored at different temperatures for six months (Ziegler et al., 2018b)
Rice	13%	6 months	16, 24, 32 and 40 °C	Effects of storage temperature of whole rice grains with brown, black and red pericarps, on the physicochemical and pasting properties (Ziegler et al., 2017)
Rice	12.5%, 16%, 19% and 21%	16 weeks	10, 15, 20, 27 and 40 °C	Impacts of storage temperature and rice moisture content on color characteristics of rice from fields with different disease management practices (Shafiekhani et al., 2018)
Rice	13%	6 months	16, 24, 32 and 40 °C	Pigmented rice oil: Changes in oxidative stability and bioactive compounds during storage of whole grains (Ziegler et al., 2017c)
Maize	13%	12 months	5, 15, 25 and 35 °C	Quality of maize grains stored at different temperatures (Paraginski et al., 2015)
Maize	13%	12 months	5, 15, 25 and 35 °C	Physicochemical and pasting properties of maize as affected by storage temperature (Paraginski et al., 2014)
Maize	13%	12 months	5, 15, 25 and 35 °C	Characteristics of starch isolated from maize as a function of grain storage temperature (Paraginski et al., 2014b)
Maize	<14%	6 months	22 °C	Carotenoid retention in biofortified maize using different post-harvest storage and packaging methods (Taleon et al., 2017)
Maize	13.5%–14.5%	6 months	–	Comparative effects of hermetic and traditional storage devices on maize grain: Mycotoxin development, insect infestation and grain quality (Walker et al., 2014)
Maize	14%, 16%, 18% and 20%	60 days	27 °C	Impact of moisture content and maize weevils on maize quality during hermetic and non-hermetic storage (Suleiman et al., 2018)
Canola	8%, 10%, 12% and 14%	20 weeks	10, 20, 30 and 40 °C	Quality changes in high and low oil content canola during storage: Part I e Safe storage time under constant temperatures (Sun et al., 2014)
Rye	10.0%, 12.5%, 15.0% and 17.5%	16 weeks	10, 20, 30 and 40 °C	Comparison of deterioration of rye under two different storage regimes (Rajarammanna and White, 2010)
Oats	11%–14%	12 and 24 months	25 °C	The impact of storage on the primary and secondary metabolites, antioxidant activity and digestibility of oat grains (Avena sativa) (Racik et al., 2014)
Lentil	10%, 12% and 14%	360 days	15, 25 and 35 °C	Multivariate analysis of the conditions of temperature, moisture and storage time in the technological, chemical, nutritional parameters and phytochemical of green lentils (Bragança et al., 2020)
Lentil	10%, 12.5%, 15.0% and 17.5%	16 weeks	10, 20, 30 and 40 °C	Effect of storage conditions on red lentils (Sravanthi et al., 2013)

4.1. Changes in physiological and biochemical properties of grains as a function of storage

Carioca beans stored at 12 °C, maintained the germination index unchanged for 240 days of storage, regardless of the humidity. However, at 36 °C, and at moisture levels of 13.8% and 16.7%, the germination index and vigor of the grains reduced to zero at 80 days of storage (Demito et al., 2019). Likewise, other studies have proven the detrimental effect of high grain moisture, storage time, and temperature on germination in pinto beans (Rani et al., 2013), lentils (Sravanthi et al., 2013), wheat (Zhang et al., 2017), maize (Paraginski et al., 2015), canola (Sun et al., 2014), soy (Smaniotto et al., 2014), and rye (Rajarammanna and White, 2010). The low storage temperatures reduce the respiration rate of the grains (Suleiman et al., 2018), in addition to maintaining the viability of enzymes involved in the germination process, which helps in maintaining the physiological properties of the grains.

4.2. Changes in chemical composition and digestibility of starch in grains as a function of storage

The changes in chemical composition observed during storage are basically due to the metabolism of grains and associated microorganisms. In black beans, Ferreira et al. (2017) found no changes ($P \geq 0.05$) in the composition of the beans due to time (12 months), temperature (11, 18, 25, and 32 °C) and moisture (14% and 17%) during storage. On the other hand, Rani et al. (2013) found a reduction ($P < 0.05$) in the protein content of the pinto beans in storage after 16 weeks, at 20% humidity, regardless of the storage temperature used. In soybeans, Ziegler et al. (2016b) found a reduction in protein content from 37.92% to 35.10%, when the grains were stored at 16% of moisture, and at a temperature of 25 °C, for 12 months.

The evidence of protein reduction in soybeans during storage was also verified by Lee and Cho (2012) and by Liu et al. (2008), with these reductions being potentiated due to high temperatures and high humidity during storage. The content of lipids in soybeans is also altered depending on storage conditions, as demonstrated by Ziegler et al. (2016), who found a reduction of approximately 3% in the lipid content in soybeans stored at 16% of moisture and at 25 °C. Rakic et al. (2014) found an average reduction of 10.98% and 15.98% in the lipid content, 7.55% and 11.82% in the protein content, 6.65% and 14.77% in the crude cellulose content, and 6.48% and 9.48% in the starch content of three oat cultivars, at 12 and 24 months of storage respectively, at 25 °C. In rice grains, with brown, black, and red pericarp, Ziegler et al. (2017) found no changes ($P \geq 0.05$) in the content of proteins, lipids, and ash, as a function of temperature (16, 24, 32, and 40 °C) and time (6 months). Paraginski et al. (2014) found no changes ($P \geq 0.05$) in the basic composition of maize grains stored at an initial moisture of 14%, depending on the temperature (5, 15, 25, and 35 °C) and storage time (12 months).

It is generally observed that grains with less fat content (rice, beans, and maize), have better stability during storage, when compared to grains with a higher fat content (soy and oats), probably due to the higher water activity present in oilseeds, which is associated with a greater reactivity of lipid compounds. These factors result in a greater respiratory activity of the grains and oxidative degradation of the lipids, which leads to a reduction of proteins and lipids in the grains. In the case of soybeans, these two basic components have high added value and directly affect the production and derivation of products such as refined oil, soybean meal, and isolated protein. Hence, adequate storage conditions, aimed at maintenance of these components, are important.

Another parameter for evaluating the quality of grains is the digestibility of starch, which for grain storage is a recent issue. In carioca beans, Demito et al. (2019) studied starch digestibility as a function of moisture (13.8% and 16.7%), temperature (12, 20, 28, and 36 °C), and storage time (240 days), and found that the starch digestibility went from 57.35% (13.8% moisture) and 58.04% (16.7% moisture) at the beginning of storage, to 8.37% (13.8% moisture) and 10.22% (16.7% moisture) in grains stored for 9 months, at 36 °C; while in storage at 12 °C, after 240 days, digestibility was 43.84% and 43.04%, at 13.8% and 16.7% of moisture, respectively. In oat grains stored with moisture between 11% and 14%, at a temperature of 25 °C, Racík et al. (2014) found a reduction in the digestibility of organic matter from 75.57% to 69.62% in cultivar Vrbas and from 75.01% to 71.11% in cultivar NS Tara, after 24 months of storage. The study of digestibility of starch, depending on storage conditions, is relevant, as it affects the nutritional use of these grains, both for human consumption and consumption by animals as feed. The study of digestibility of starch in other grains depending on storage conditions, characterization of their behaviors, and a probe of ways to minimize the negative effects resulting from the storage conditions, is suggested.

4.3. Changes in bioactive properties of grains as a function of storage

The behavior of bioactive compounds, more specifically of phenolic compounds in grains during storage, is still not well established. However, there are reports that the temperature-humidity-microorganism interaction can stimulate the synthesis of phenolic compounds in grains during storage. In oat grains, Rakic et al. (2014) found a reduction of 28.79% and 41.19% in the content of total phenolics and caffeic acid, respectively, after 24 months of storage at 25 °C. Reduction of bioactive substances was also observed in black beans by Ferreira et al. (2017), in storage under different conditions of temperature (11, 18, 25, and 32 °C) and grain moisture (14% and 17%) over 12 months, with the most significant reduction being seen in conditions of higher humidity and temperature. In general, the storage period of grains reduces their concentration of bioactive compounds, especially in adverse conditions. However, some exceptions can be observed, as in the study with soybeans carried out by Ziegler et al. (2016), where there was a significant increase in total phenolics and flavonoids, in soybeans with moisture content of 12, 15, and 18%, especially at the highest study temperatures (25 °C and 30 °C) after 12 months of storage.

4.4. Changes in technological properties as a function of storage

The process of cooling the grain mass, during the storage period, is an effective and economical technique for maintaining the quality of the grains, as it decreases the water activity and the respiration rate of the grains, and also delays the development of insects, pests, and microflora (Ziegler et al., 2016). Several studies carried out to evaluate the effect of temperature on storage of grains concluded that this is a factor that affects the physicochemical properties of the grains, increasing the acidity of the lipids, the color and hardness of the grains, and changing the sensory and texturometer properties of the grains after cooking (Ziegler et al., 2016; Demito et al., 2019; Ferreira et al., 2018).

The reduction of moisture content of the grains for storage, carried out via the drying process, helps in the conservation of the grains. It reduces the water activity and consequently the speed of biochemical and enzymatic reactions in the grains, thus helping in the prevention of fungal development. When artificial cooling techniques are not used, the moisture content of grains recommended for storage, is 12% for rice and soy, and 13%, for wheat, beans, and maize.

The storage period for grains can vary according to the species and time of the year. The longer this period, the greater is the care needed with moisture content and temperature of the grains. It should be noted that monitoring and evaluating these characteristics over time is essential for monitoring the quality and possible commercialization of these grains. For example, maize stored at 14% moisture and at 25 °C, with 6 months of storage, already they were out of type (Paraginski et al., 2015).

The relationship between humidity, temperature, and storage time are fundamental in determining safe storage conditions for grains, thus ensuring maintenance of the bioactive, technological, and biological parameters of grains of different species. The use of adequate humidity, combined with inadequate temperature and time, can compromise the quality of the grains.

5. Conclusions

Grain storage systems are essential for obtaining good quality grains which are safe for consumption. Therefore, correct management of grains with simple practices must always be adopted, to help in the maintenance of the characteristics of the grains for long periods. These practices seek to improve the current condition of the field the postharvest. Several systems are available for storage and these are affected by the grain moisture, temperature, and storage time for each species. Specific studies of each of these factors need to be considered for decision making, as they will assist in the reduction of qualitative and quantitative losses of grains.

Credit author statement

Valmor Ziegler: Conceptualization, Writing- Reviewing and Editing. **Ricardo Tadeu Paraginski:** Conceptualization, Writing-Reviewing and Editing. **Cristiano Dietrich Ferreira:** Writing-Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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